

FATIGUE AND FRACTURE MECHANICS
PROFESSIONAL ELECTIVE-I

VI Semester										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A5AE39		PEC		L	T	P	C	CIE	SEE	Total
				3	0	0	3	40	60	100
UNIT-I	FATIGUE OF STRUCTURES									
Introduction to fatigue, Endurance limits - Effect of mean stress, Goodman, Gerber and Soderberg relations and diagrams - Notches and stress concentrations - Neuber's stress concentration factors - Plastic stress concentration factors - Notched S.N. Curves.										
UNIT-II	PHYSICAL ASPECTS OF FATIGUE									
Phase in fatigue life - Crack initiation - Crack growth - Final fracture - Dislocations - Fatigue fracture surfaces.										
UNIT-III	STATISTICAL ASPECTS OF FATIGUE BEHAVIOUR									
Low cycle and high cycle fatigue - Coffin - Manson's relation - Transition life - Cyclic strain hardening and softening - Analysis of load histories - Cycle counting techniques -Cumulative damage - Miner's theory.										
UNIT-IV	OVERVIEW OF ENGINEERING FRACTURE MECHANICS									
Strength of cracked bodies - Potential energy and surface energy - Griffith's theory - Irwin - Orwin extension of Griffith's theory to ductile materials - Stress analysis of cracked bodies - Effect of thickness on fracture toughness.										
UNIT-V	FATIGUE DESIGN AND TESTING									
Safe life and fail-safe design philosophies, Importance of fracture mechanics in aerospace structure, Application to structures										
Text Books:										
1. J. F. Knott (1983), Fundamentals of Fracture Mechanics, Butter Worth & Co., Publishers Ltd., London.										
Reference Books:										
1. Prashant Kumar, "Elements of Fracture Mechanics", Tata McGraw Hill, New Delhi, India, 2009.										
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:										
1. "Ability to use simple continuum mechanics and elasticity to determine the stresses, strains, and displacements in a loaded structure."										
2. Understanding and mathematical modeling of the elements of plastic deformation, with respect to continuum and microscopic mechanisms.										
3. "Ability to use creep data to predict the life of structures at elevated temperatures and the understanding of mechanisms of creep deformation and fracture"										
4. "Use of fracture mechanics to quantitatively estimate failure criteria for both elastically and plastically deforming structures, in the design of life prediction strategies, and for fracture control plans, with examples from automotive, aerospace, medical, and other industries."										
5. Understanding of fatigue and how this affects structural lifetimes of components										